

Assignment # 6

Builder Journal

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Internship: AI Engineering Fellowship 2026

Organization: Invictus Solutions

Introduction:

This builder journal summarizes my learning experience during the first week of the AI Engineering Fellowship 2026. It reflects my motivation for choosing AI Engineering, the projects I worked on, the challenges I faced, and the goals I have set for the upcoming week.

1. Why Did You Choose AI Engineering?

I chose AI Engineering because Artificial Intelligence is transforming the way software is built and used. I have always been interested in software development, but I wanted to work on technologies that can solve real-world problems through automation and intelligent decision-making.

AI Engineering combines programming, machine learning, prompt engineering, and automation into one field. I believe these skills will help me build innovative applications and prepare me for future career opportunities. My goal is to become an AI Engineer capable of creating practical AI-powered solutions that improve productivity and simplify everyday tasks.

2. What Type of AI Products Do You Want to Build?

I want to develop AI-powered applications that are useful in real-life situations. Some of the products I hope to build include:

- AI chat assistants for businesses
- AI email automation systems
- AI customer support agents
- AI workflow automation using n8n
- Educational AI assistants for students
- Intelligent business automation systems

I am particularly interested in combining Large Language Models (LLMs) with automation platforms to create practical AI solutions.

3. What Was Your Biggest Learning This Week?

This week introduced me to many important AI Engineering concepts.

Some of my biggest learnings were:

- Understanding Prompt Engineering techniques.
- Learning different prompting methods such as Role Prompting, Few-Shot Prompting, Structured Output, and Prompt Optimization.

- Building my first AI Workspace application using Lovable.
- Learning how AI applications are structured through architecture diagrams.
- Publishing projects and reports on GitHub.
- Learning the basics of AI workflow automation using n8n.

These experiences helped me understand that AI Engineering is not only about using AI models but also about designing complete software systems around them.

4. What Challenges Did You Face?

During this week, I faced several technical challenges.

The biggest challenge was integrating a real AI API into my AI Workspace application. Although I successfully generated a Google AI Studio API key, I encountered compatibility issues because the application expected a different API key format.

I also experienced difficulties understanding project deployment, GitHub repositories, publishing applications, and managing project files. Since these technologies were new to me, learning them required patience and continuous practice.

5. How Did You Solve Them?

I solved these challenges by researching documentation, asking questions, and practicing repeatedly.

I learned how GitHub repositories work, how to upload reports, organize project files, and publish my AI application. Although the API integration is still under development, I now understand how API keys, environment variables, and backend integration work together.

Instead of giving up when I encountered errors, I treated them as learning opportunities. Each problem improved my understanding of AI application development.

6. What Are Your Goals for Week 2?

For the second week of the internship, I have set the following goals:

- Successfully connect my AI Workspace with a real AI model using an API.
- Learn more about AI automation using n8n.
- Build practical AI workflow projects
- Learn backend integration for AI applications.
- Strengthen my GitHub skills by maintaining organized project repositories.
- I aim to gain practical experience and build real-world AI solutions.